DESIGN SYSTEMS

TECHNICAL DATA SHEET PAVIFLOOR A+B
EPOXY FORMULATION IN AQUEOUS EMULSION

DESCRIPTION

PAVIFLOOR A+B is a two-component finish based on water-dispersed epoxy and amine resins, with a high solids content, for coatings with high chemical resistance, ideal for floors such as marble-effect ResinFloor.

- Excellent wear resistance.
- Easy cleaning and sanitising of the surface.
- Good resistance to water, alkaline solutions, detergents and hydrocarbons, both high-boiling and low-boiling (oil, diesel, petrol, etc.).

PAVIFLOOR is suitable for coating floors and walls in rooms where food is present, subject to HACCP system assessments in accordance with Reg. (EC) 852/2004 (Italian Legislative Decree 193/2007).

The formulation is supplied pre-coloured for some shades or in a non-pigmented (NP) version, with water-based colour paste supplied separately.

USES

Coloured painting of concrete floors inside buildings. Finish for epoxy-cement or high-build epoxy coatings. Suitable for food environments, as it is easy to clean and sanitise. Colour regeneration of resin floors. Coating for skirting in aseptic rooms in the food and pharmaceutical industries, in hospitals and in public areas.

SUBSTRATE

The substrate must have a minimum compressive strength of 25 N/mm^2 and a tensile strength of 1,5 N/mm^2.

SUBSTRATE PREPARATION

Concrete substrates must be solid, dry (cured if newly laid), level, absorbent, not contaminated by oils, detergents, release agents, dust or other substances. Assess the most suitable type of mechanical preparation (abrading or grinding). Any holes and minor defects can be repaired beforehand with epoxy filler. Tiled floors must be abraded or ground until a completely matt surface is obtained.

Existing resin coatings must be abraded or ground, then the dust residues removed. Absorbent cementitious surfaces or those tending to dust must first be treated with a water-based epoxy primer or with a coat of PAVIFLOOR diluted with 50% water by weight. Substrates prepared by grinding that show abrasion marks must be skim-coated with suitable formulations. Damp substrates or substrates subject to capillary rising damp must be treated with a specific primer (contact the Design Systems technical service).

APPLICATION

Pre-coloured PAVIFLOOR

At the time of application, combine the two pre-dosed components in a single container. Mix carefully for 2 minutes using suitable equipment (drill with mixing paddle), taking care to clean the sides of the containers

with a spatula so as to mix all the product. Once mixing is complete, respect the induction time* by letting the product rest in the tin. Slowly add the dilution water while stirring (recommended 20% for the first coat, 10% for the finishing coat), mix again for 1 minute and apply by roller, at a consumption of 0,130 kg/m^2 per coat.

PAVIFLOOR NP + colour paste

Stir the colour paste well to homogenise any sediment. At the time of application, add the colour paste to component A and mix carefully. Then add component B and mix for 2 minutes using suitable equipment (drill with mixing paddle), taking care to clean the sides of the containers with a spatula. Once mixing is complete, respect the induction time* by letting the product rest in the tin. Slowly add the dilution water while stirring (recommended 20% for the first coat, 10% for the finishing coat), mix again for 1 minute and apply by roller, at a consumption of 0,130 kg/m^2 per coat.

To obtain an anti-slip surface, add 3-5% by weight of GLASS BEADS (FINE or COARSE), keeping the product stirred to prevent settling. PAVIFLOOR can also be applied as a skim coat: in this case add 30% by weight of COARSE GLASS BEADS to the mix and apply with a steel finishing trowel, for a maximum consumption of 0,200 kg/m^2 of PAVIFLOOR.

* PAVIFLOOR must be applied after the induction time and within the usable time, depending on the ambient temperature (see technical data).

WARNINGS

Avoid preparing partial mixes of product, so as not to incur accidental errors that could lead to failed or incomplete curing. If the induction and usable times are not respected, the finish may be uneven, showing differences in colour and degree of gloss. As a finish it must be applied by roller, evening out the surface with parallel movements.

For applications at low temperatures, the formulation can be diluted with a 1:1 mixture of water and ethyl alcohol. Protect the applied product from humidity, condensation and water for at least 24 hours.

Yellow, orange and some red shades may require more coats to obtain good hiding power (in some cases an additional first coat in white is recommended). Different production batches of the same colour may show small differences: where possible, use material from a single batch for each area to be treated.

Some colours based on organic pigments (reds, blues, greens, intense yellows, etc.) tend to release colour when subjected to abrasion, dry or wet: in these cases it is advisable to protect the colour with a coat of clear finish. Do not apply on gypsum-based plasters intended for damp rooms (e.g. showers).

PRODUCT DATA

DESCRIPTION	RESULT
Colour	On request according to the colour chart
Consumption	0,130 kg/m^2 per coat

Specific gravity (at 25°C)	A+B mix: 1,48 ± 0,05 g/ml; Part A: 1,55 ± 0,05 g/ml; Part B: 1,10 ± 0,05 g/ml
Viscosity (at 25°C)	A+B mix: 3.500-4.000 mPa · s (spindle 2, 5 rpm); A+B mix diluted at 10%: 800-1.300 mPa · s (spindle 2, 5 rpm)
Solids content (A+B)	69% by weight, 54% by volume
VOC ready for use (Italian Legislative Decree 161/06)	< 50 g/l – Cat. A/j: two-component high-performance coating (BA)
Flash point	Not applicable
Tool cleaning	Water
Storage	12 months, in a dry place at a temperature between 10°C and 35°C

APPLICATION DATA AND TIMES

DESCRIPTION	RESULT
Mixing ratio (by weight)	Pre-coloured version: A = 100, B = 28; NP version: A = 85, paste = 15, B = 28
Induction time	10 min (30°C) – 15 min (25°C) – 25 min (15°C)
Pot life (50% RH)	at 15°C: 130 min; at 25°C: 105 min; at 30°C: > 80 min
Touch dry (50% RH)	at 15°C: 18-22 hours; at 25°C: 4-6 hours; at 30°C: 2,5-3,5 hours

DESCRIPTION	RESULT
Foot traffic (50% RH)	at 25°C: from 14 to 36 hours
Overcoating (50% RH)	at 25°C: from 14 to 36 hours
Full traffic (50% RH)	at 25°C: 3 days
Full curing (50% RH)	at 25°C: 7 days
Ambient conditions for use	Temperature between +15°C and +35°C, RH < 70%
Surface temperature	Between +10°C and +30°C
Maintenance of the coating	Use neutral detergents for cleaning

TECHNICAL PERFORMANCE DATA

DESCRIPTION	RESULT
Appearance	Slightly orange-peel surface
Gloss at 60°	60-65 (*)
Abrasion resistance (UNI 8298-9)	55 mg (Taber, CS-17 wheel, 1000 cycles, 1000 g)
Abrasion resistance (UNI EN ISO 5470-1)	84 mg
Pull-off adhesion (EN 1542)	3,6 MPa

Impact resistance (EN ISO 6272-1)	11,8 N · m (class 2)
Water permeability (EN 1062-3)	w = 0,012 kg/(m² · h^0,5)
Reaction to fire (UNI EN 13501-1)	Class Bfl-s1
Slip resistance (Italian Ministerial Decree 236/89)	μ > 0,40 (with 3% FINE GLASS BEADS)
Chemical resistance	Contact the Design Systems technical service

(*) The degree of gloss of the applied PAVIFLOOR is influenced by several factors: material temperature, ambient humidity, application temperature and porosity of the substrate.

The technical data given in this bulletin were obtained using standardised tests and standards, or modifications thereof, according to Design Systems systems. The methods applied may be requested from our technical service. The contents of this technical bulletin represent the best of our knowledge and experience. However, for uses other than those indicated, preliminary tests are recommended. In any case, Design Systems Srl accepts no responsibility for damage or defects caused by the use of its products, since the conditions of use are not under its control. It also advises that its technical service is available to customers for any information concerning the correct use of its products.